FAWAD ARSHAD

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PROFESSIONAL EXPERIENCE

SOFTWARE DEVELOPER (Co-op)

Lead Manager CA *January 2026 – April 2026*

Flutter · Dart · TypeScript · Firebase · HTML · Shell scripts

- **Developed an AI-powered SMS sales agent** using Twilio and OpenAI LLMs with shopper profiles. Automated usage throttling and token limits to prevent API overages.
- **Developed a secure Vehicle Management System** compliant with OMVIC, automating garage registry tracking
- **Implemented server-side gating and usage tracking** for SMS, and API actions via Firebase Cloud Functions, enabling tier-based access control for B2B clients.
- **Automated testing and deployment workflows using CI/CD,** ensuring tests are executed prior to each deployment.

TECHNICAL CONSULTANT (Co-op)

Genesis Jewelry Boutique *May 2025 – August 2025*

- **Spearheaded digitization of a retail boutique's legacy inventory architecture**, integrating Square POS with WooCommerce for multi-channel sales.
- **Formulated a comprehensive technical ROI assessment** projecting a 10–15% sales increase by eliminating inventory errors and establishing a centralized inventory ledger.
- **Engineered cloud-migration strategy** that automated legacy workflows, reducing daily manual inventory adjustments by approximately 80%.

PROJECTS

FRAGMENTS REST API

JavaScript · Node.js · Express · AWS (S3, DynamoDB) · Docker · Cognito

- **Auto deployed on version tags via GitHub Actions** (Jest/Hurl tests → ECR → ECS) using a CI/CD pipeline, with LocalStack for offline AWS testing

IN-MEMORY KEY-VALUE STORE

C++ · TCP Sockets · GDB · Multiplexing I/O · RESP Protocol

- **Event-driven in-memory key-value store** supporting concurrent TCP client connections via POSIX sockets and I/O multiplexing, using unordered_map for O(1) data lookups. Redis Serialization Protocol used to handle byte streams.

END-TO-END AUTONOMOUS VEHICLE CONTROL CNN

TensorFlow · OpenCV · Python · Pandas · Scikit-learn

- **Used the Freedman–Diaconis rule to calculate histogram bin widths**, downsampling straight-line frames by 60%.

ENTERPRISE INVENTORY BACKEND SERVICE

Java · Spring Boot · JDBC · SQL · Maven

- **Built a Java and Spring Boot backend system** to manage records securely and handle simultaneous user requests

SKILLS

Languages: Java, C++, Python, JavaScript, SQL, PLSQL, TypeScript, HTML, CSS

Web & App Development: React, React Native, Next.js, Expo, Spring Boot, Django, Express.js, Node.js, Flutter, Selenium

Databases & Cloud: MongoDB, Firebase, AWS, DynamoDB

Tools & DevOps: Docker, Git, CI/CD

EDUCATION

HONOURS BACHELOR OF TECHNOLOGY – SOFTWARE DEVELOPMENT (Co-op)

Seneca Polytechnic | *Sept 2023 - December 2026 (Expected Graduation)*

COMPUTER PROGRAMMING DIPLOMA

Humber College | *January 2022 – August 2023*